

RMIT University
School of Computing Technologies
COSC2667 / COSC2777 — Data Science and AI Postgraduate Projects
Assessment 1: Project Proposal Report

300,000 Streets - School Streets

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Student Team

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Executive Summary

Regen Melbourne's 300,000 Streets initiative seeks to improve street safety and liveability across Greater Melbourne. This project, conducted as part of RMIT's COSC2667/2777 Data Science and AI Postgraduate Projects course, investigates street conditions surrounding schools in high-growth suburban corridors where rapid development has outpaced pedestrian infrastructure investment. Using geospatial techniques — including spatial buffer analysis, kernel density estimation, and network analysis the project will assess streets within 400m and 800m of three to five Melbourne schools, integrating datasets on road networks, crash records, and socio-economic context to classify hazards by severity. Deliverables include a geospatial database, spatial visualisations, a composite Street Safety Score per school, and a prioritized recommendation report to support Regen Melbourne's engagement with local councils. All work will be completed over ten weeks using Python, QGIS, and open-source spatial libraries.

1. Partner / Client and Background

1.1 About Regen Melbourne

Regen Melbourne is a not-for-profit urban regeneration organisation based in Melbourne, Victoria, whose primary mission is to reimagine urban spaces so that they are more liveable, sustainable, and community-centred. The organisation works collaboratively with local councils, state government agencies, and community groups to drive structural improvements to how Melbourne's streets, neighborhoods, and public spaces are designed and experienced.

Through its 300,000 Streets initiative, Regen Melbourne is undertaking a city-wide programme to evaluate and improve the condition of streets across Greater Melbourne. The initiative is grounded in the recognition that streets are not merely transport infrastructure they are critical shared assets that shape public health, community cohesion, environmental sustainability, and equitable access to urban life. The scale of the programme its name derived from the estimated number of street segments across the metropolitan area reflects Regen Melbourne's ambition to establish a replicable, data-driven framework for street-level urban improvement.

1.2 The Business Problem

Regen Melbourne has identified school-adjacent streets as a priority concern within the 300,000 Streets initiative. Across Greater Melbourne — particularly in rapidly expanding corridors including Wyndham, Melton, and the City of Casey — streets surrounding primary and secondary schools frequently lack the infrastructure needed for safe active travel. Common deficiencies include absent or deteriorated footpaths, inadequate pedestrian crossings, insufficient lighting, and speed limits incompatible with the safety needs of children during school hours. These conditions generate quantifiable safety risks and deter walking and cycling, pushing families toward private vehicle use. The consequences include traffic congestion around school gates, elevated local air pollution, reduced physical activity among students, and greater danger for those who do walk or cycle — outcomes that extend beyond individual safety to broader public health, environmental, and urban planning concerns. Regen Melbourne currently lacks a comprehensive, spatially referenced assessment of street safety conditions around schools. Existing evaluations have been limited in scope, reliant on isolated manual observations, and insufficient to support prioritised, evidence-based advocacy to councils and government agencies. This collaborative research project directly addresses that gap.

1.3 Why Data Science and Geospatial Analysis Can Help

The scale of the problem school zones distributed across hundreds of suburbs in a metropolitan area of more than 9,900 square kilometres, renders comprehensive manual assessment logistically and financially infeasible. Data science methods provide a principled means of processing, integrating, and analysing large volumes of heterogeneous spatial data at a scale and resolution that manual review cannot achieve.

Specifically, geospatial data science enables the systematic identification of infrastructure deficiencies across road networks, the detection of spatial clusters of hazardous conditions through statistical techniques such as kernel density estimation, the quantification of safety attributes through composite scoring, and the integration of socio-economic context through data linkage. These capabilities allow the project to move beyond anecdotal or reactive assessment towards a reproducible, evidence-based methodology that can be applied consistently across multiple school zones and, ultimately, scaled to the wider 300,000 Streets programme.

1.4 Existing Systems and Related Work

Several relevant initiatives provide context for this project. The Victorian Government's School Safety Infrastructure Program funds improvements to school zones, but decisions are largely reactive and not systematically driven by comprehensive spatial analysis. VicRoads maintains road network and speed zone datasets, and the Transport Accident Commission (TAC) publishes crash statistics, but neither organisation produces integrated school-specific pedestrian safety assessments. Internationally, programmes such as Safe Routes to School (United States) and Living Streets (United Kingdom) have demonstrated the value of data-driven approaches to school street safety, though no directly equivalent framework exists for Melbourne's growth corridors (Safe Routes to School National Partnership, 2023; Living Streets, 2022). This project draws on these precedents while adapting the methodology to the specific data landscape and planning context of Victoria.

1.5 Key Terms and Definitions

The following terms are used consistently throughout this proposal and are defined here for clarity:

- **Active Travel** — Any form of human-powered travel, principally walking, cycling, and wheeling. In the school context, this refers to students and families travelling to and from school without a motor vehicle.
- **School Zone** — A designated area surrounding a school within which reduced speed limits apply during specified hours. In Victoria, school zones operate at 40 km/h during pick-up and drop-off times on school days (VicRoads, 2023).
- **Buffer Zone** — A spatial area defined by a set radius from a point of interest, generated through GIS analysis. This project uses 400m (approximately a five-minute walk) and 800m (approximately a ten-minute walk) buffer zones around each selected school.
- **SEIFA** — Socio-Economic Indexes for Areas, a suite of indices produced by the Australian Bureau of Statistics (ABS) to summarise the relative socio-economic advantage and disadvantage of geographic areas, based on Census data (ABS, 2021).
- **OpenStreetMap (OSM)** — A freely available, collaboratively maintained global geographic database used in this project as a primary source of road network, footpath, and crossing data (OpenStreetMap Contributors, 2024).
- **Kernel Density Estimation (KDE)** — A non-parametric statistical technique used to estimate the probability density function of a spatial point pattern, employed in this project to identify clusters of crash incidents and infrastructure deficiencies.
- **GDA2020** — Geocentric Datum of Australia 2020, the current national standard coordinate reference system for spatial data in Australia, to which all project datasets will be standardised (ICSM, 2020).

1.6 Project Stakeholders

The following stakeholders have been identified as having a direct or material interest in the outcomes of this project. Their inclusion in the project record reflects best practice in professional project management and ensures that the scope and communication plan remain aligned with the full set of parties whose interests this work serves.

Stakeholder	Role	Description
Regen Melbourne	Industry Partner & Primary Beneficiary	Commissioning organisation. Will use project outputs recommendations and spatial data as advocacy evidence to present to councils and government agencies on behalf of the 300,000 Streets initiative.
Darebin City Council	Local Government Authority	Key implementing authority for the study area. Responsible for local road and footpath infrastructure decisions in Reservoir and Preston. Has conducted the <i>Your Street, Your Say</i> community consultation relevant to this project.
Victorian Department of Transport and Planning	State Government Infrastructure Decision Maker	Oversees state-level road infrastructure funding programmes, including the School Safety Infrastructure Program and the Victorian Black Spot Programme. A key audience for Regen Melbourne's advocacy work arising from this project.
School Communities (Reservoir High School, William Ruthven Secondary College, Preston High School)	Direct Beneficiaries	Students, families, and staff at the three selected schools are the direct beneficiaries of any infrastructure improvements recommended by this project. They are also the community most directly affected by existing street safety deficiencies.
Transport Accident Commission (TAC)	Government Agency Data Provider	Custodian of the Victorian Crash Statistics dataset used in this project. Has an organisational interest in the reduction of pedestrian road trauma, particularly in school zones.
RMIT University	Academic Supervisor & Assessment Body	Provides academic oversight through the COSC2667/2777 course framework. Responsible for assessing project deliverables and ensuring research integrity.

The primary audience for all project outputs is Regen Melbourne as the commissioning partner. However, the project team acknowledges that the ultimate impact of the work depends on uptake by Darebin City Council and the Victorian Department of Transport and Planning, both of whom will be indirect audiences for the recommendation outputs. This awareness informs the requirement that all recommendations be spatially precise, costed, and referenced to specific hazard records, the level of evidence required to initiate formal council and government infrastructure review processes.

2. Project Aims

2.1 Primary Aim

The primary aim of this project is to conduct a comprehensive, reproducible, data-driven geospatial analysis of street conditions surrounding selected schools in Greater Melbourne, in order to assess safety, accessibility, and suitability for active travel, and to generate prioritised, evidence-based recommendations for infrastructure improvement.

2.2 Minimum (Baseline) Aims

The following baseline objectives represent the minimum set of outcomes that the team commits to delivering. These aims are designed to be achievable within the ten-week project timeline even under conditions of data or resource constraint. They correspond directly to the core deliverables agreed with the industry partner:

- Identify and justify the selection of three to five schools in Greater Melbourne growth corridors as the spatial focus for the analysis.
- Collect, clean, and integrate a minimum of three publicly available datasets covering road networks, footpath infrastructure, and pedestrian crash or hazard records within defined school buffer zones.

- Generate 400m and 800m buffer zones around each selected school using Python (GeoPandas) or QGIS.
- Identify and spatially locate infrastructure deficiencies and safety hazards within each buffer zone, classified by type and severity.
- Produce clear spatial visualisations maps and heatmaps, illustrating the distribution of identified hazards for each school zone.
- Develop a minimum of three prioritised, evidence-based recommendations per school, directly linked to identified hazards and ranked by severity and feasibility.

2.3 Desirable (Extended) Aims

The following extended objectives will be pursued if the project progresses ahead of schedule or if data availability exceeds current estimates. These aims represent additional analytical and communicative value but are not required for successful project completion:

- Incorporate environmental data layers — specifically air quality indices and noise exposure estimates — to enrich the safety assessment with factors beyond physical infrastructure.
- Develop a composite Street Safety Score for each school zone, integrating footpath quality, crossing safety, traffic volume, and environmental exposure into a single quantitative indicator suitable for cross-school comparison.
- Conduct a systematic virtual field observation programme using Google Street View imagery to validate and supplement findings from dataset-derived analysis.
- Produce an interactive web map or dashboard enabling Regen Melbourne to explore findings dynamically, filtered by school, hazard type, or recommendation priority.
- Investigate whether a statistically significant correlation exists between SEIFA disadvantage scores and street infrastructure quality within school catchment areas, testing the hypothesis that students in lower-SEIFA areas experience disproportionately inferior street conditions.

2.4 Measurability of Aims

Each aim above is designed to be measurable. Success against the baseline aims will be assessed by: (a) the existence and completeness of the geospatial database, (b) the spatial accuracy and legend completeness of produced maps, (c) the traceability of each recommendation to a specific, severity-rated hazard record, and (d) the reproducibility of the analysis pipeline from raw data inputs to final outputs. Extended aims will be assessed against the additional metrics defined in the deliverables table in Section 3.8.

2.5 Alignment with Industry Partner Priorities

The aims above were developed in consultation with Regen Melbourne's 300,000 Streets initiative objectives. The industry partner has confirmed that its primary requirement is a set of spatially referenced, prioritised recommendations that can be presented to local councils and state government bodies as advocacy evidence for targeted infrastructure investment. All project aims are structured to serve this requirement directly. The tiered aims structure baseline and extended reflects the Agile principle of delivering value incrementally, ensuring that a useful and complete output is achieved even if extended goals are not fully realised within the project timeframe.

3. Methodology and Expected Deliverables

3.1 Overall Methodological Approach

This project adopts an iterative, Agile-inspired geospatial data science methodology, structured across five sequential but mutually informing phases. The Agile approach is particularly appropriate here because the industry partner has acknowledged uncertainty about the completeness of available data and the precise scope of feasible analytical goals. Rather than committing to a rigid linear plan, the team will conduct regular sprint reviews at the conclusion of each two-week period, using findings from each phase to refine

the scope and approach of subsequent phases without compromising delivery of the core baseline aims. The five phases are: Phase 1 — Data Collection and Scoping; Phase 2 — Data Preprocessing and Integration; Phase 3 — Feature Engineering and Hazard Identification; Phase 4 — Spatial Analysis and Visualisation; and Phase 5 — Recommendation Development. Each phase is described in detail below, with explicit justification for the methodological choices made.

3.2 Phase 1 — Data Collection and Scoping

This phase involves the systematic identification, sourcing, and initial validation of all datasets required for the analysis. The following publicly available data sources will be used: Victorian Department of Education School Register (school locations, type, and enrolment); OpenStreetMap via the Overpass API (road network geometry, footpath presence, crossing locations, and speed zone tags); VicRoads Open Data Portal (speed limits and road classifications); Victorian Crash Statistics Dataset (geocoded pedestrian crash records, 2012–2022); ABS 2021 Census SEIFA indices at SA1 scale; EPA Victoria air quality data where available; and Google Street View for manual validation of infrastructure conditions at flagged locations.

All datasets are publicly available, contain no personally identifiable information, and no ethical approval is required. The most significant data quality risk is incomplete OSM footpath coverage in outer suburban growth corridors, mitigated through the virtual field observation approach described above.

3.3 Phase 2 — Data Preprocessing and Integration

Raw datasets from heterogeneous sources require substantial preprocessing before spatial analysis can begin. Key steps include: reprojection to GDA2020/MGA Zone 55 (EPSG:7855) for accurate distance-based operations; attribute schema harmonisation across OSM and VicRoads classifications; spatial joins linking road segments, footpaths, and crossings to school buffer zones; missing value treatment via median imputation or explicit 'Unknown' flagging; and construction of a structured geospatial database with seven linked tables — Schools, Buffers, Street Segments, Footpaths, Crossings, Hazards, and Recommendations. All data will be stored as portable GeoPackage files compatible across Python and QGIS environments.

3.4 Phase 3 — Feature Engineering and Hazard Identification

This phase derives analytical features enabling quantitative comparison of street safety conditions across all school zones. Three composite indicators will be constructed:

- Footpath Accessibility Score (FAS, 0–10): weighted composite of footpath presence (0.4), width relative to the 1.5m minimum standard (0.2), surface condition (0.2), kerb ramp availability (0.1), and continuity (0.1).
- Crossing Safety Score (CSS, 0–10): derived from crossing type (signalised = 10 through to unmarked = 0), sightline obstruction, and distance from the nearest school gate.
- Environmental Exposure Indicator (EEI, 0–10): combining mean PM2.5 concentration during school hours (0.6) and estimated noise exposure from traffic volume and speed (0.4).

Hazards will be classified as Minor (score 7–10), Moderate (score 4–6.9), or Major (score below 4, or any unprotected location within 100m of a school gate). Identification combines rule-based detection with kernel density estimation of crash clusters using a 200m bandwidth validated against local road network structure using OSMnx.

3.5 Phase 4 — Spatial Analysis and Visualisation

Four analytical methods will be applied, each selected for their validity in pedestrian safety research and availability in open-source Python libraries:

- Buffer Analysis (GeoPandas): 400m and 800m Euclidean buffers generated around each school gate, representing standard five- and ten-minute walking catchments per Australian urban planning guidance.
- Network Analysis (OSMnx): walkable street network graphs constructed per school catchment; circuitry indices calculated to assess pedestrian route directness, following Boeing (2017).
- Kernel Density Estimation (SciPy): Gaussian KDE with a 200m bandwidth applied to crash and infrastructure deficiency points, producing smoothed density surfaces classified into quintiles and

visualised as choropleth heatmaps overlaid on the street network.

- **Overlay Analysis (GeoPandas):** infrastructure quality, crash density, and SEIFA layers combined to identify locations where poor infrastructure, elevated crash risk, and socio-economic disadvantage converge — enabling prioritisation where safety needs are greatest.

All analysis will be conducted using Python 3.11 and QGIS 3.34 on standard laptops. No licensed software, cloud computing, or specialised hardware is required.

3.6 Phase 5 — Recommendation Development

The final phase translates findings into actionable, spatially referenced recommendations for use by Regen Melbourne in its engagement with councils and government agencies. Each Major hazard — and selected Moderate hazards in high-priority locations — will generate a structured recommendation record containing: intervention type (a specific, actionable description of the proposed infrastructure change); priority level (High, Medium, or Low) based on hazard severity, proximity to the school gate, and estimated daily student exposure; estimated cost band (Low: under \$20,000; Medium: \$20,000–\$200,000; High: above \$200,000) referencing Victorian Local Government Association unit cost benchmarks; implementation timeframe (short- or long-term); and precise spatial coordinates enabling direct use by council engineering and planning teams.

3.2 Potential Risks, Challenges, and Mitigations

Risk	Likelihood	Impact	Mitigation Strategy
Incomplete OSM footpath coverage in outer suburban growth corridors	High	High	Supplement with Google Street View virtual observation; cross-reference with VicRoads road feature data; document and disclose all data gaps in the final report
Pedestrian crash dataset lacks sufficient spatial precision for school-level filtering	Medium	Medium	Apply 400m spatial filter around school gates; cross-reference with TAC crash maps; treat low-precision records as a data limitation and document accordingly
No physical access to school sites for field observation	High	Low	Google Street View provides sufficient visual validation for the purposes of this project; all virtual observations to be documented with timestamps and image references
EPA air quality monitoring coverage insufficient near selected schools	Medium	Low	Use traffic volume multiplied by speed limit as a proxy noise/pollution indicator; disclose the proxy in all outputs
OSMnx network graph construction fails for sparse suburban road networks	Low	Medium	Fall back to Euclidean buffer analysis for affected schools; document the limitation; test network construction during Week 3 to identify issues early
Team member availability disrupted by competing academic commitments	Low	Medium	Weekly milestone check-ins; task redistribution protocol agreed in advance; all code documented to enable knowledge transfer between members
Inconsistent CRS across datasets causes spatial misalignment	Low	High	All datasets to be reprojected to GDA2020/MGA55 as the first step of preprocessing; CRS validation check after every spatial join operation

3.3 Expected Deliverables and Measurable Success Criteria

The following deliverables will be produced by the conclusion of the project. Each is defined with explicit success criteria to ensure measurability and to demonstrate contribution towards the project aims:

Deliverable	Description	Measurable Success Criteria
Geospatial Database	A structured, linked relational database comprising seven tables (Schools, Buffers, Street Segments, Footpaths, Crossings, Hazards, Recommendations) in GeoPackage format	All seven tables populated with validated data; no null primary keys; all spatial joins confirmed accurate to within 1m in QGIS; database schema documented
School Zone Infrastructure Maps	A4 cartographic maps for each selected school showing buffer zones, road network, footpath presence/absence, crossing locations, and hazard point overlays	One map produced per school; includes scale bar, north arrow, legend, and coordinate grid; reviewed and approved by all team members prior to submission
Hazard Heatmaps	KDE-based spatial density maps showing the concentration of crash incidents and infrastructure deficiencies within each school buffer zone	Produced for all selected schools; bandwidth and classification method documented; hotspot areas clearly distinguishable at A4 print resolution
Street Safety Score Summary	A tabular and visual summary of Footpath Accessibility Score, Crossing Safety Score, and Environmental Exposure Indicator for each school zone, enabling cross-school comparison	Scores calculated and documented for all selected schools; scoring methodology fully disclosed; scores reproducible from the underlying database
Prioritised Recommendation Report	A spatially referenced table and accompanying map of recommended infrastructure interventions, ranked by priority level and estimated cost	Minimum three Major/High-priority recommendations per school; each recommendation linked to a specific hazard record by ID; spatial geometry included for all recommendations
Final Analytical Report	A comprehensive project report documenting background, methodology, findings, limitations, and recommendations in accordance with RMIT assessment requirements	Submitted by the Week 10 deadline; all sections complete; reviewed by all team members; bibliography formatted in APA 7th edition

4. Project Plan

4.1 Approach to Planning and Project Management

The team will operate under an Agile sprint methodology, structured around two-week development cycles. At the beginning of each sprint, the team will review progress against the deliverables defined in Section 3.8, identify any blockers or scope adjustments required, and assign tasks for the forthcoming fortnight. A standing weekly check-in of approximately 30 minutes will be held every Monday via Microsoft Teams to maintain momentum between sprint boundaries. All code, analytical notebooks, and project documentation will be maintained in a shared private GitHub repository. Each team member will work on a dedicated named branch; all contributions will be merged to the main branch via pull requests requiring at least one peer review and approval. This workflow ensures code quality, provides an auditable record of individual contributions, and enables rapid identification and resolution of integration issues. The project task board will be maintained in GitHub Projects, using a Kanban-style board with columns for Backlog, In Progress, In Review, and Done. Each task will be assigned to a team member, tagged with the relevant project phase, and linked to the corresponding deliverable. This provides a transparent, real-time record of individual contributions that can be referenced in weekly engagement records.

4.2 Week-by-Week Timeline with Responsibilities and Dependencies

Week	Sprint Goal & Tasks	Responsible Member(s)	Key Dependency
Week 1 (Initiation)	Kick-off meeting with Regen Melbourne; confirm school selection criteria; review RMIT assessment rubric; establish GitHub repository and Teams channel; draft data collection plan; agree on team protocols and conflict resolution procedure	All members (Sameer: project coordinator)	None
Week 2 (Scoping)	Conduct targeted literature review on school street safety and active travel in Australian context; finalise selection of 3–5 schools with written justification referencing SEIFA, growth corridor status, and data availability; test API access for OSM Overpass and EPA	Pranay (data scoping), Sameer (literature, school justification)	Week 1: kick-off complete, schools provisionally agreed
Week 3 (Data Collection)	Download and catalogue all required datasets; assess coverage and quality against data quality checklist; document format, CRS, update frequency, and known limitations for each dataset; flag any coverage gaps for early mitigation	Pranay & Nagendra (data collection); Hishikesh (QGIS environment setup)	Week 2: school selection finalised
Week 4 (Preprocessing)	Reproject all datasets to GDA2020/MGA55; harmonise attribute schemas; conduct spatial joins to associate features with school buffer zones; generate 400m and 800m buffers; handle missing values; construct initial GeoPackage database	Nagendra (preprocessing pipeline), Siddhartha (spatial joins & buffer generation)	Week 3: all datasets collected and catalogued
Week 5 (Feature Engineering)	Implement Footpath Accessibility Score, Crossing Safety Score, and Environmental Exposure Indicator calculations; conduct internal peer review of scoring logic and weighting rationale; write feature engineering documentation	Siddhartha (score computation), Pranay (weighting review), Sameer (documentation)	Week 4: preprocessing and database construction complete
Week 6 (Hazard ID)	Apply rule-based hazard detection logic; run KDE clustering on crash and deficiency point data; classify all identified hazards by severity level; conduct Google Street View virtual validation for a sample of Major-severity hazards; populate Hazards table in database	Siddhartha & Pranay (hazard identification and classification)	Week 5: feature scores calculated and validated
Week 7 (Spatial Analysis)	Conduct network analysis using OSMnx for each school zone; perform multi-layer overlay analysis; produce draft hazard point maps and KDE heatmaps for all schools in QGIS; circulate draft maps for team review	Siddhartha (OSMnx, overlay), Hishikesh (QGIS map drafts)	Week 6: hazards identified and classified

Week 8 (Recommendations)	Develop structured recommendation records for all Major hazards and priority Moderate hazards; assign priority levels, cost levels, and timeframes; populate Recommendations table; draft recommendation narrative for report	Sameer (recommendation narrative), Nagendra (database population)	Week 7: spatial analysis complete
Week 9 (Visualisation)	Finalise all QGIS cartographic maps to submission standard (scale bars, legends, north arrows); produce composite safety score summary tables and charts; begin interactive web map if ahead of schedule; prepare all figures for report integration	Hishikesh (cartography and charts), Pranay (interactive map if applicable)	Weeks 7–8 complete

4.3 Inter-Member Dependencies and Interaction Points

The following specific interaction points between team members have been identified as critical to maintaining schedule integrity:

Weeks 3–4 handoff (Pranay/Nagendra → Siddhartha): Pranay and Nagendra must deliver a complete, catalogued, and CRS-standardised dataset package to Siddhartha by the end of Week 4. Any delays in data collection will directly delay feature engineering.

Week 5–6 peer review (Siddhartha → Pranay): the scoring logic for all composite indicators will be peer-reviewed by Pranay before being applied to the full dataset. This interaction point ensures methodological accuracy and provides a second set of eyes on all weighting decisions.

Week 7 map review (Hishikesh → All): draft QGIS maps will be circulated to all team members for review against the defined map success criteria (scale bar, legend, legibility) before finalisation.

Week 8–9 data freeze: by the end of Week 8, all database tables must be considered final. Hishikesh will begin final visualisation production in Week 9 using fixed data; any changes to the database after this point require approval from Sameer as project coordinator.

4.4 Agile Contingency Planning

Two specific contingency plans have been developed in advance:

Data gap contingency (Weeks 3–4): if a required dataset is found to be unavailable or of insufficient quality, the team will convene an unscheduled check-in within 48 hours to agree on an alternative source or proxy approach. The affected deliverable's success criteria will be updated to reflect the data limitation, and the limitation will be explicitly disclosed in the final report.

Scope reduction contingency (Week 7 sprint review): if analysis of all five selected schools cannot be completed within the available time, the team will reduce the scope to the three highest-priority schools (those with the lowest SEIFA scores or the largest proportion of students walking to school) and document the scope reduction transparently in the final report.

5 Team Roles, Responsibilities, and Collaboration

5.2 Individual Roles and Responsibilities

Each team member has been assigned a primary role aligned to the project's analytical phases. All members are additionally expected to contribute to internal peer review, weekly check-in attendance, and the final report. Role assignments were agreed by the team at the Week 1 kick-off meeting based on individual skill sets, prior experience, and availability. No member's primary role precludes them from contributing to other phases where capacity allows.

Team Member	Primary Role	Phase Responsibility	Secondary Contributions
Vipparla Reddy Pranay	Data Lead & GIS Analyst	Phases 1, 3	Dataset API integration; GeoPandas buffer generation; peer review of scoring logic; interactive map (Phase 4, if applicable)
Venkata Nagendra Anamala	Data Engineer	Phases 2, 5	CRS standardisation and reprojection; attribute schema harmonisation; spatial joins; GeoPackage database construction; recommendation database population.
Siddhartha Ananthula	Spatial Analyst & Modeller	Phases 3, 4	Feature score computation; rule-based hazard detection; KDE clustering; OSMnx network analysis; overlay analysis; primary hazard identification
Hishikesh Phukan	Visualisation Specialist	Phases 4, 5	QGIS cartographic map production; KDE heatmap design; composite score charts; QGIS project file management; final figure preparation for report
Sameer Yadav	Project Coordinator & Lead Author	All phases	Weekly agenda preparation; milestone tracking; GitHub Projects board maintenance; recommendation narrative; report integration and consistency review; industry partner liaison

5.3 Collaboration Tools and Protocols

GitHub (private repository) All Python scripts, Jupyter notebooks, QGIS project files, and written documentation will be stored under version control. Named branches per member; mandatory peer review via pull requests before merging to main. Repository URL to be shared with academic supervisor upon request. Microsoft Teams — Primary communication platform. Weekly 30-minute check-in every Monday at an agreed time. All decisions and action items documented in the meeting notes channel. File sharing for non-code assets (e.g., draft report sections). GitHub Projects (Kanban board) — Task tracking across all project phases. Each card assigned to a team member, tagged with phase and priority, and linked to a GitHub issue for traceability. Shared Google Drive — Collaborative drafting of report sections, meeting notes, and the dataset documentation register. Access shared with academic supervisor. QGIS Shared Project Files — Stored on Google Drive for team access. All cartographic layer files and map compositions maintained in a single QGIS project file, version-labelled.

5.4 Conflict Resolution Protocol

In the event of a disagreement about methodology, scope, or task allocation, the following structured resolution process will be applied:

Step 1: the team member who identifies the disagreement raises it at the next weekly Monday check-in, or calls an unscheduled meeting if the issue is time-sensitive.

Step 2: each party states their position clearly; the group discusses the evidence and implications.

Step 3: the team reaches a decision by majority vote; in the event of a tied vote, Sameer as project coordinator has the casting vote.

Step 4: if the disagreement cannot be resolved internally within 48 hours of the check-in, the matter is escalated to the academic supervisor with a written summary of both positions.

All decisions, regardless of how resolved, are documented in the meeting notes on Microsoft Teams and the Google Drive meeting log.

The team additionally agrees that individual contributions to the GitHub repository constitute the primary record of each member's work and will be referenced in weekly engagement and contribution records. All members commit to making their contributions visible, documented, and reviewable by teammates.

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